

DAY 17

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DATE: 29-08-2025

Task:

- Call the favouriteBookAPI from Angular and display the values in the form
 - user can update the name, price etc., in the form
- When submit button is pressed, call the updateBOOK api from angular
- (optional) include the token via interceptor when calling the update api
- (optional) also put the delete button, when clicked call the delete api

Code:

bookentry.interface.ts

```
export interface Author
{
  authorId:number,
  authorName:string
}

export interface NewBook{
  newBookId: number,
  title: string,
  price: number,
  authorId: number,
  author: Author,
  no_Of_Sold: number
}
```

book.service.ts

```
import { Injectable } from '@angular/core';
import { HttpClient, HttpHeaders } from '@angular/common/http';
import { Author, BookEntry, FetchBook, NewBook } from 'src/models/bookentry.interface';
import { User } from 'src/models/user.interface';
```

```

@Injectables({
  providedIn: 'root'
})
export class BookService {

  private apiUrl = "http://localhost:5241/Book";
  private apiUrl_user = "http://localhost:5241/User";
  constructor(private http:HttpClient) { }
  private admin_usr: Usr = {name: "Jimson", password: "pass"}

  getToken(){
    this.http.post(`${this.apiUrl_user}/Login`, this.admin_usr, { responseType: 'text' }).subscribe(
      value => {
        console.log(`Token: ${value}`)
        localStorage.setItem('token', value)
      }
    )
  }

  AddNewBook(bookentry: BookEntry)
  {
    const book_url = `${this.apiUrl}/AddNewBook`
    return this.http.post(book_url, bookentry, { responseType: 'text' })
  }

  RemoveNewBook(book_id:number)
  {
    const book_url = `${this.apiUrl}/RemoveNewBook?id=${book_id}`
    return this.http.delete(book_url, { responseType: 'text' })
  }

  UpdateNewBook(book : NewBook)
  {
    const book_url = `${this.apiUrl}/UpdateNewBook`
    return this.http.post(book_url, book, { responseType: 'text' })
  }

  fetchAllNewBooks(){
    return this.http.get<NewBook[]>(`${this.apiUrl}/FetchAllNewBooks`)
  }

  fetchAuthors() {
    return this.http.get<Author[]>(`${this.apiUrl}/FetchAuthors`)
  }
}

```

samp.component.ts

```
import { Component } from '@angular/core';
import { BookService } from '../book.service';
import { Author, BookEntry, FetchBook, NewBook } from 'src/models/bookentry.interface';

@Component({
  selector: 'app-samp',
  templateUrl: './samp.component.html',
  styleUrls: ['./samp.component.scss']
})
export class SampComponent {

  AuthName:string = ""
  BookName:string = ""
  Cost:number= 0
  Copies:number= 0
  new_book: BookEntry = {authorName:"", bookTitle:"", price:0, no_Of_Sold:0}
  editingBook: NewBook | null = null;
  fetch_book:NewBook[] = []
  authors: Author[] = [];

  addBook()
  {
    this.new_book = {authorName: this.AuthName, bookTitle: this.BookName, price: this.Cost,
no_Of_Sold: this.Copies}

    this.bookService.AddNewBook(this.new_book).subscribe(
      value => {
        console.log(`Response from server: ${value}`)
      }
    )
    console.log("after calling the add api")
  }

  listBook()
  {
    this.bookService.fetchAllNewBooks().subscribe(
      value => {
        this.fetch_book = value
      }
    )
  }
}
```

```

deleteBook(bookId: number)
{
  this.bookService.RemoveNewBook(bookId).subscribe(
    value => {
      console.log(`Response from server: ${value}`)
      this.listBook();
    }
  )
}

editBook(book: NewBook) {
  this.editingBook = { ...book };
}

cancelEdit() {
  this.editingBook = null;
}

saveBook() {
  if (this.editingBook) {
    this.bookService.UpdateNewBook(this.editingBook).subscribe(
      res => {
        alert('Book updated successfully');
        this.editingBook = null;
        this.listBook();
      },
      err => console.error("Error:", err)
    );
  }
}

loadAuthors()
{
  this.bookService.fetchAuthors().subscribe(
    data => {
      this.authors = data;
    }
  )
}

```

author.interceptor.ts

```
import { Injectable } from '@angular/core';
import {
  HttpRequest,
  HttpHandler,
  HttpEvent,
  HttpInterceptor
} from '@angular/common/http';
import { Observable } from 'rxjs';
import { Token } from '@angular/compiler';

@Injectable()
export class AuthorInterceptor implements HttpInterceptor {

  constructor() {}

  intercept(request: HttpRequest<unknown>, next: HttpHandler):
  Observable<HttpEvent<unknown>> {
    // debugger
    const token_stored = localStorage.getItem('token')
    request = request.clone({headers: request.headers.set('Authorization', 'bearer '+
token_stored)})
    console.log("Added the JWT token")
    return next.handle(request);
  }
}
```

samp.component.html

```
<h2>Book Entry Form</h2>
<form>
  Author: <input type="text" name="auth" [(ngModel)]="AuthName" placeholder="Enter book
author name"><br/><br/>
  Book Name: <input type="text" name="name" [(ngModel)]="BookName" placeholder="Enter
book name"><br/><br/>
  Price: <input type="number" name="price" [(ngModel)]="Cost" placeholder="Enter price of the
book"><br/><br/>
  Number of Copies: <input type="number" name="copy" [(ngModel)]="Copies"
placeholder="Enter no of copies sold"><br/><br/>
</form>
<button (click)="addBook()">Add Book</button><br/><br/>
```

```

<h5>-----</h5>
<button (click)="listBook()">List Book</button>
<h3>List of Books</h3>
<div *ngFor="let book of fetch_book">
  <p>
    <b>Book Name:</b> {{book.title}} - <b>Author:</b> {{book.author.authorName}} -
    <b>Price:</b> Rs.{{book.price}} - <b>Sold:</b> {{book.no_Of_Sold}} copies
    <button (click)="editBook(book)">Edit</button>&nbsp;
    <button (click)="deleteBook(book.newBookId)">Delete</button>
  </p>

  <div *ngIf="editingBook?.newBookId === book.newBookId" class="edit-form">
    <h3>Edit Book</h3>
    <form (ngSubmit)="saveBook()">
      <label>Title:</label>
      <input [(ngModel)]="editingBook!.title" name="title{{book.newBookId}}" required />&nbsp;

      <label>Price:</label>
      <input type="number" [(ngModel)]="editingBook!.price" name="price{{book.newBookId}}"
required />&nbsp;

      <label>Sold:</label>
      <input type="number" [(ngModel)]="editingBook!.no_Of_Sold"
name="sold{{book.newBookId}}" />&nbsp;

      <label>Author:</label>
      <select [(ngModel)]="editingBook!.authorId" name="author{{book.newBookId}}" required>
        <option *ngFor="let a of authors" [value]="a.authorId">{{a.authorName}}</option>
      </select><br><br>

      <button type="submit">Save</button>&nbsp;
      <button type="button" (click)="cancelEdit()">Cancel</button>
    </form>
  </div>
</div>

```

BookController.cs

```

[Authorize(Roles = "Admin")]
[HttpPost("UpdateNewBook")]
public IActionResult EditNewBook(NewBook updatedBook)
{
    if (updatedBook == null)

```

```

        return BadRequest("Book is null");

var existingBook = _appDbContext.NewBooks
    .Include(b => b.author)
    .FirstOrDefault(b => b.NewBookId == updatedBook.NewBookId);

if (existingBook == null)
    return NotFound("Book not found");

existingBook.Title = updatedBook.Title;
existingBook.Price = updatedBook.Price;
existingBook.No_Of_Sold = updatedBook.No_Of_Sold;

var existingAuthor = _appDbContext.Authors
    .FirstOrDefault(a => a.AuthorId == updatedBook.AuthorId);

if (existingAuthor == null)
    return BadRequest("Invalid AuthorId");

existingBook.AuthorId = existingAuthor.AuthorId;
_appDbContext.SaveChanges();
return Ok("Book updated successfully");
}

```

Output:

Before updation & deletion - Database

	NewBookId	Title	Price	AuthorId	No_Of_Sold
1	1	Harry Potter 1	300	1	50
2	2	The Room on The Roof	250	2	100
3	3	Harry Potter 2	300	1	70
4	5	Cherry Tree	200	2	80
5	6	Burmese Days	280	3	30
6	7	Essays	320	3	53
7	8	Landour Days	330	2	10
8	16	Animal Farm	280	3	20
9	17	The Casual Vacancy	330	1	38
10	19	Troubled Blood	399	1	50
11	44	Harry Potter 4	230	1	12
12	47	Children of India	200	2	10
13	50	A nice cup of tea	199	3	20
14	53	The Hidden Pool	185	2	7

Overall UI

Book Entry Form

Author:

Book Name:

Price:

Number of Copies:

List of Books

Book Name: Harry Potter 1 - **Author:** JK Rowling - **Price:** Rs.300 - **Sold:** 50 copies

Book Name: The Room on The Roof - **Author:** Ruskin Bond - **Price:** Rs.250 - **Sold:** 100 copies

Book Name: Harry Potter 2 - **Author:** JK Rowling - **Price:** Rs.300 - **Sold:** 70 copies

Book Name: Cherry Tree - **Author:** Ruskin Bond - **Price:** Rs.200 - **Sold:** 80 copies

Book Name: Burmese Days - **Author:** George Orwell - **Price:** Rs.280 - **Sold:** 30 copies

Book Name: Essays - **Author:** George Orwell - **Price:** Rs.320 - **Sold:** 53 copies

Book Name: Landour Days - **Author:** Ruskin Bond - **Price:** Rs.330 - **Sold:** 10 copies

Book Name: Animal Farm - **Author:** George Orwell - **Price:** Rs.280 - **Sold:** 20 copies

Book Name: The Casual Vacancy - **Author:** JK Rowling - **Price:** Rs.330 - **Sold:** 38 copies

Book Name: Troubled Blood - **Author:** JK Rowling - **Price:** Rs.399 - **Sold:** 50 copies

Book Name: Harry Potter 4 - **Author:** JK Rowling - **Price:** Rs.230 - **Sold:** 12 copies

Book Name: Children of India - **Author:** Ruskin Bond - **Price:** Rs.200 - **Sold:** 10 copies

Book Name: A nice cup of tea - **Author:** George Orwell - **Price:** Rs.199 - **Sold:** 20 copies

Book Name: The Hidden Pool - **Author:** Ruskin Bond - **Price:** Rs.185 - **Sold:** 7 copies

Edit section

Book Name: Harry Potter 1 - **Author:** JK Rowling - **Price:** Rs.280 - **Sold:** 50 copies

Edit Book

Title: Price: Sold: Author:

Book Name: The Room on The Roof - **Author:** Ruskin Bond - **Price:** Rs.250 - **Sold:** 100 copies

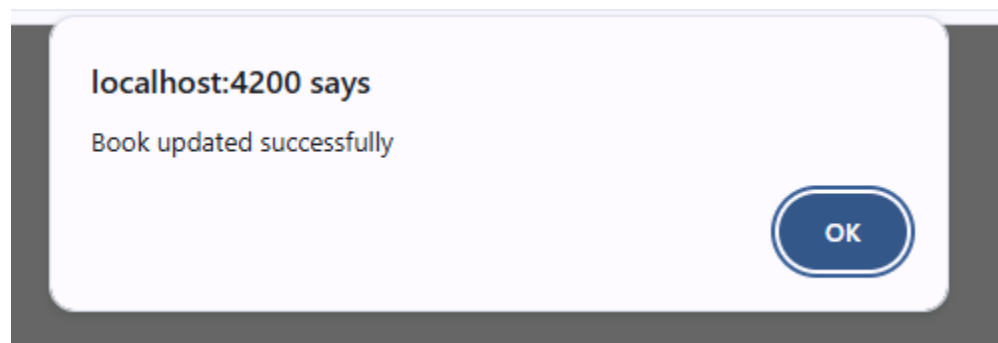
Book Name: Harry Potter 2 - **Author:** JK Rowling - **Price:** Rs.300 - **Sold:** 70 copies

After editing

Book Name: Harry Potter 1 - **Author:** JK Rowling - **Price:** Rs.300 - **Sold:** 50 copies

Edit Book

Title: Price: Sold: Author:



List of Books

Book Name: Harry Potter 1 - **Author:** JK Rowling - **Price:** Rs.280 - **Sold:** 50 copies

After updation - Database

	NewBookId	Title	Price	AuthorId	No_Of_Sold
1	1	Harry Potter 1	280	1	50
2	2	The Room on The Roof	250	2	100
3	3	Harry Potter 2	300	1	70
4	5	Cherry Tree	200	2	80
5	6	Burmese Days	280	3	30
6	7	Essays	320	3	53
7	8	Landour Days	330	2	10
8	16	Animal Farm	280	3	20
9	17	The Casual Vacancy	330	1	38
10	19	Troubled Blood	399	1	50
11	44	Harry Potter 4	230	1	12
12	47	Children of India	200	2	10
13	50	A nice cup of tea	199	3	20
14	53	The Hidden Pool	185	2	7

After deleting the cherry tree book - Response & Database

3 Added the JWT token [author.interceptor.ts:20](#)
Added the JWT token [author.interceptor.ts:20](#)
Response from server: Deleted [samp.component.ts:64](#)
Added the JWT token [author.interceptor.ts:20](#)

	NewBookId	Title	Price	AuthorId	No_Of_Sold
1	1	Harry Potter 1	280	1	50
2	2	The Room on The Roof	250	2	100
3	3	Harry Potter 2	300	1	70
4	6	Burmese Days	280	3	30
5	7	Essays	320	3	53
6	8	Landour Days	330	2	10
7	16	Animal Farm	280	3	20
8	17	The Casual Vacancy	330	1	38
9	19	Troubled Blood	399	1	50
10	44	Harry Potter 4	230	1	12
11	47	Children of India	200	2	10
12	50	A nice cup of tea	199	3	20
13	53	The Hidden Pool	185	2	7